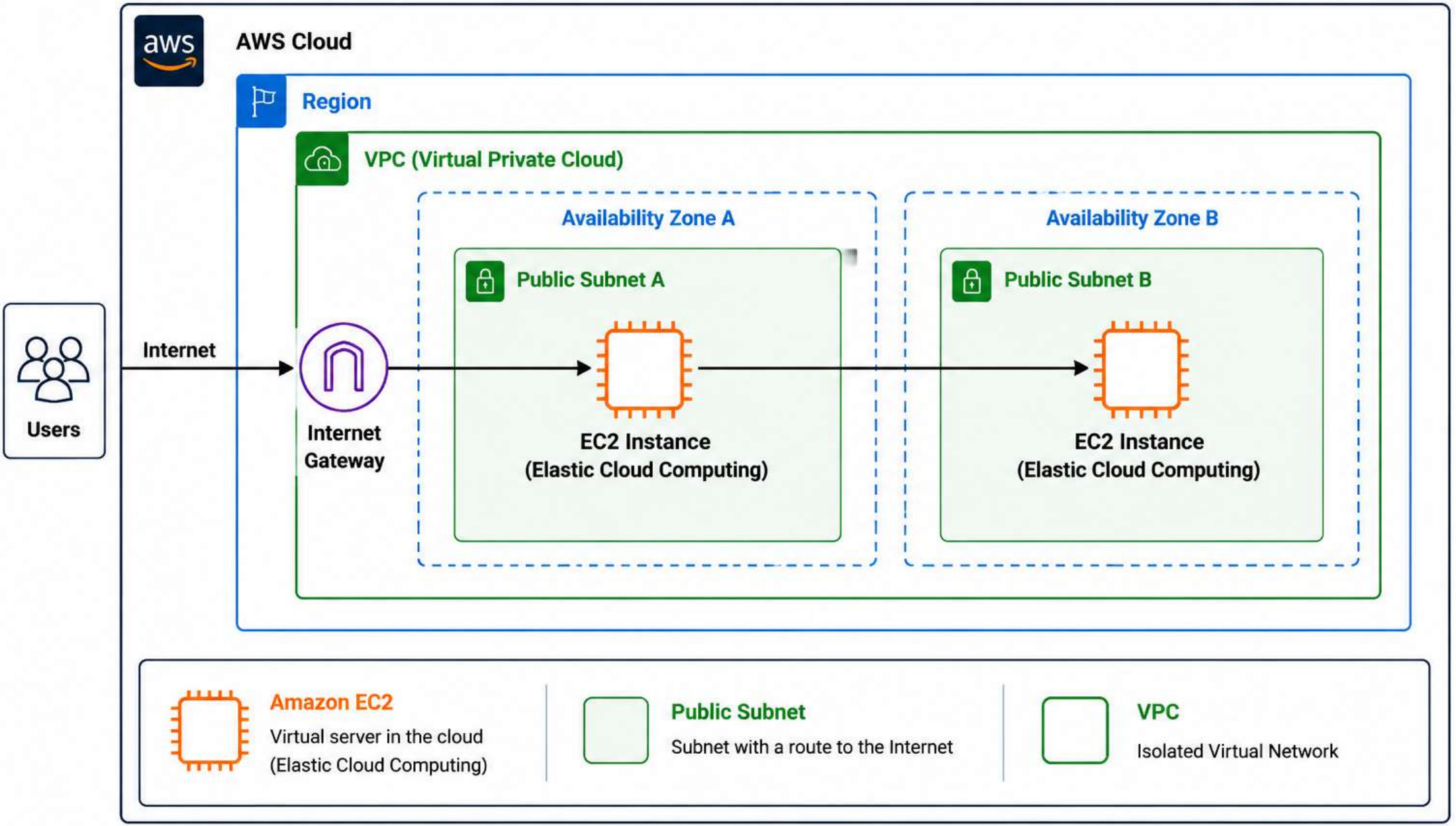
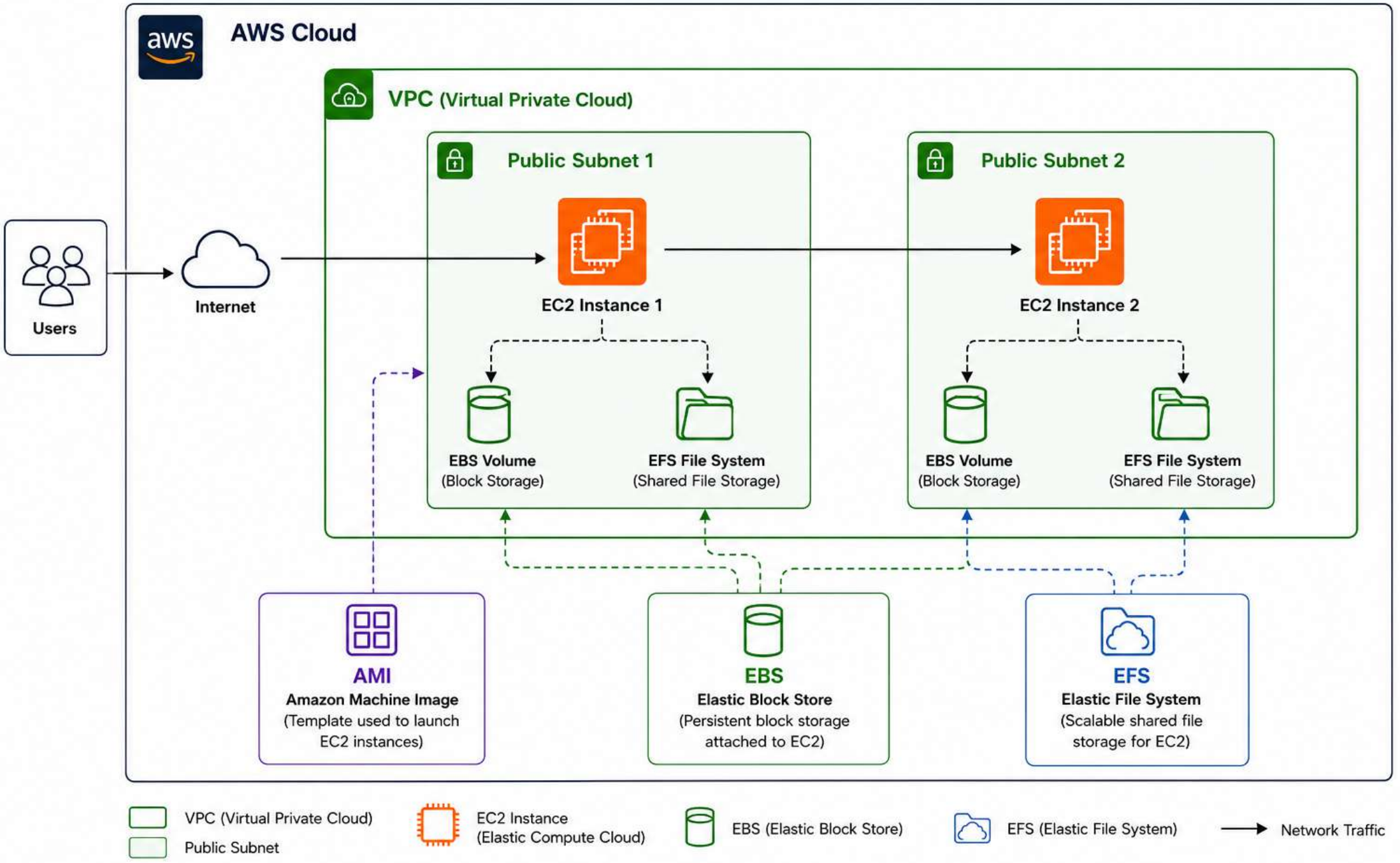
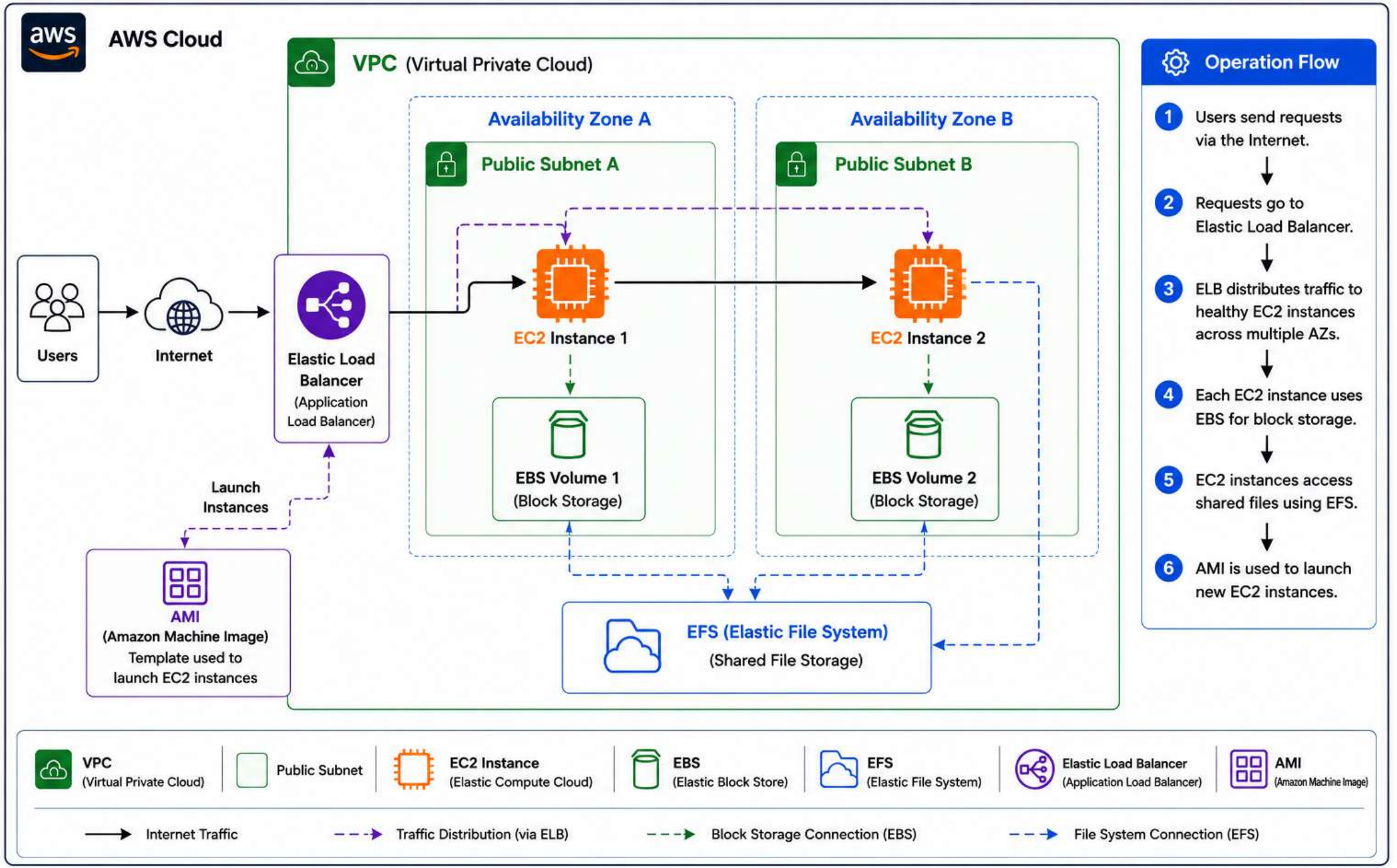
AWS EC2 with Elastic Cloud Computing



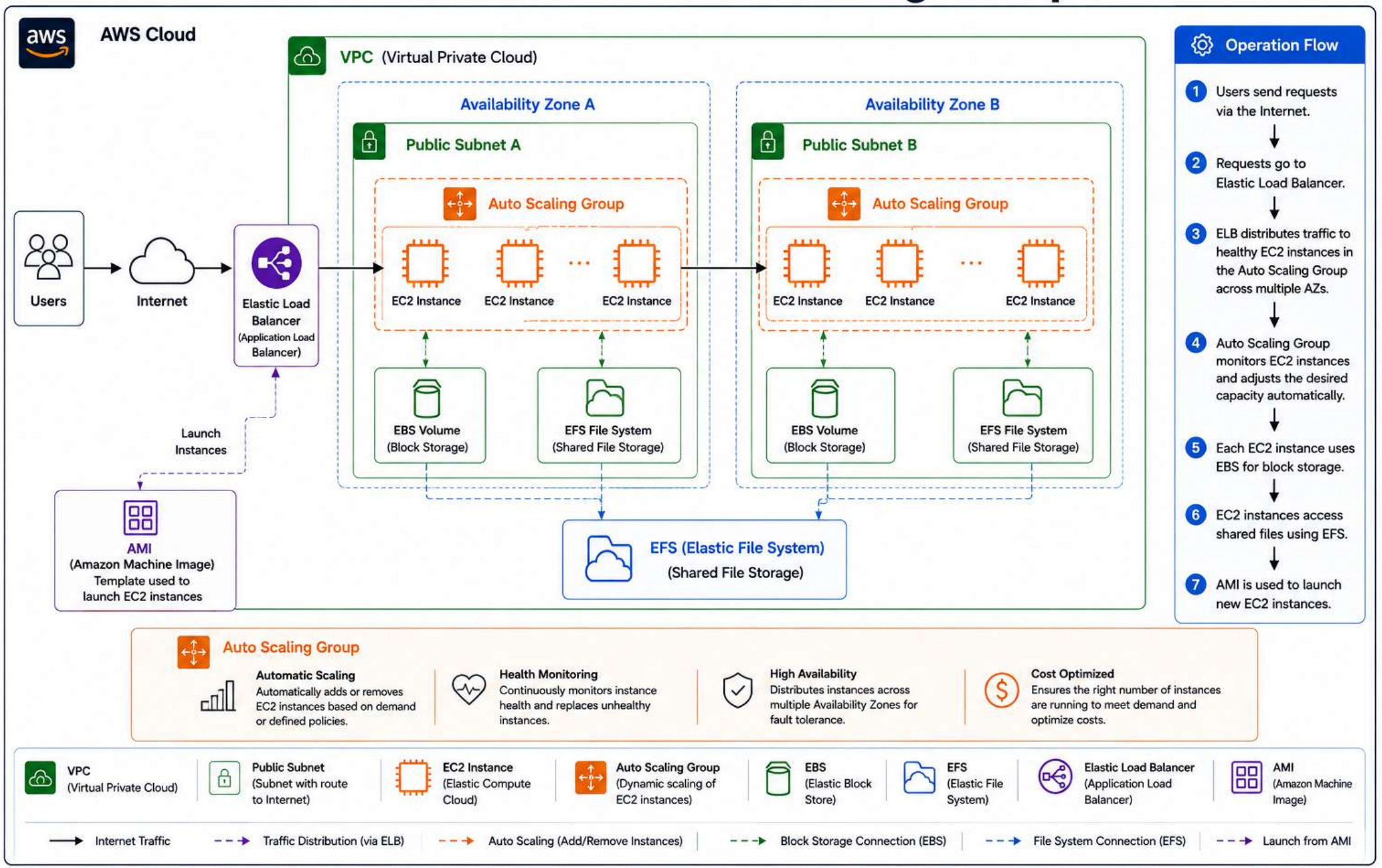
AWS EC2 with AMI, EBS & EFS



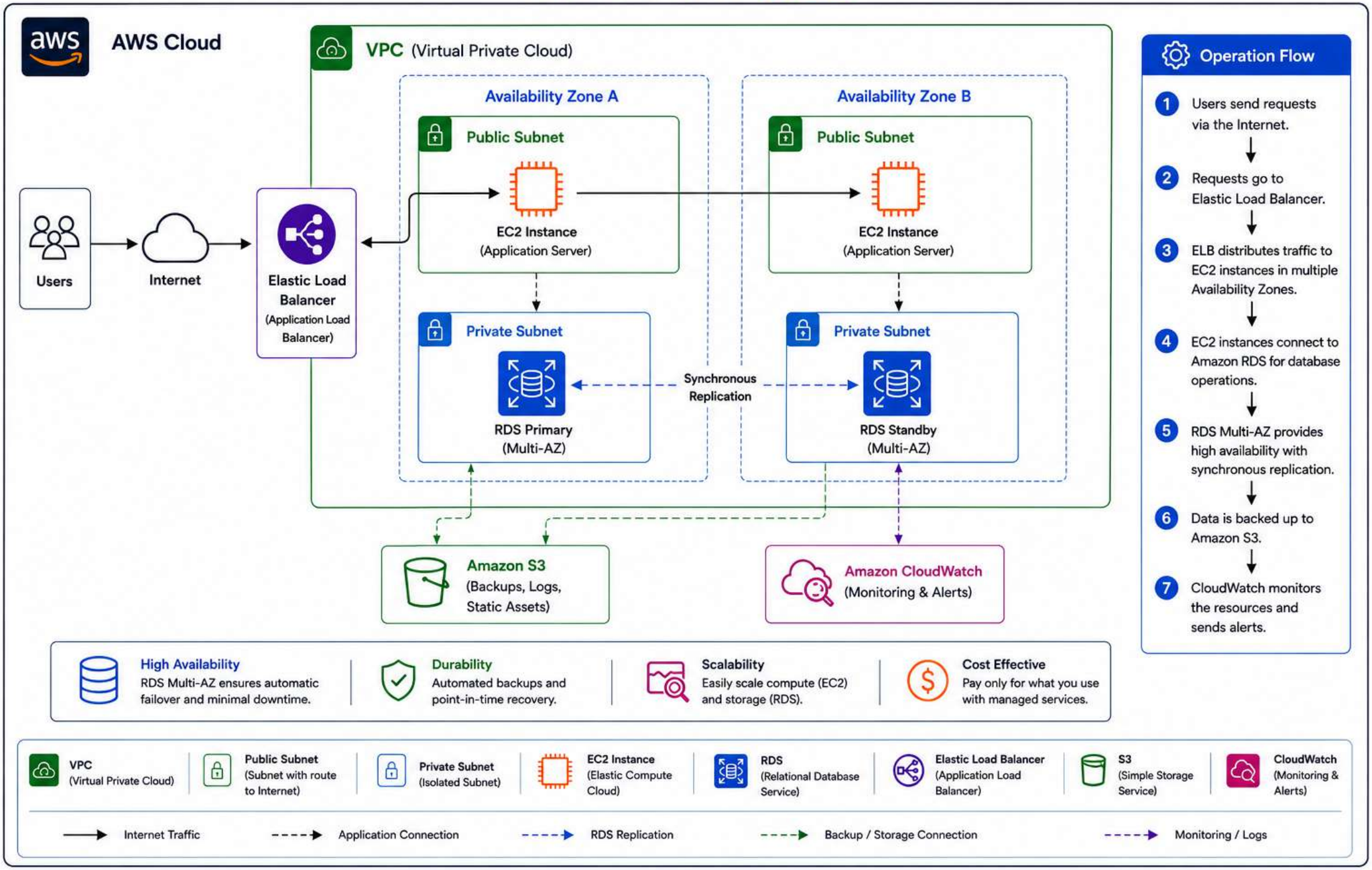
AWS EC2 with Elastic Load Balancer



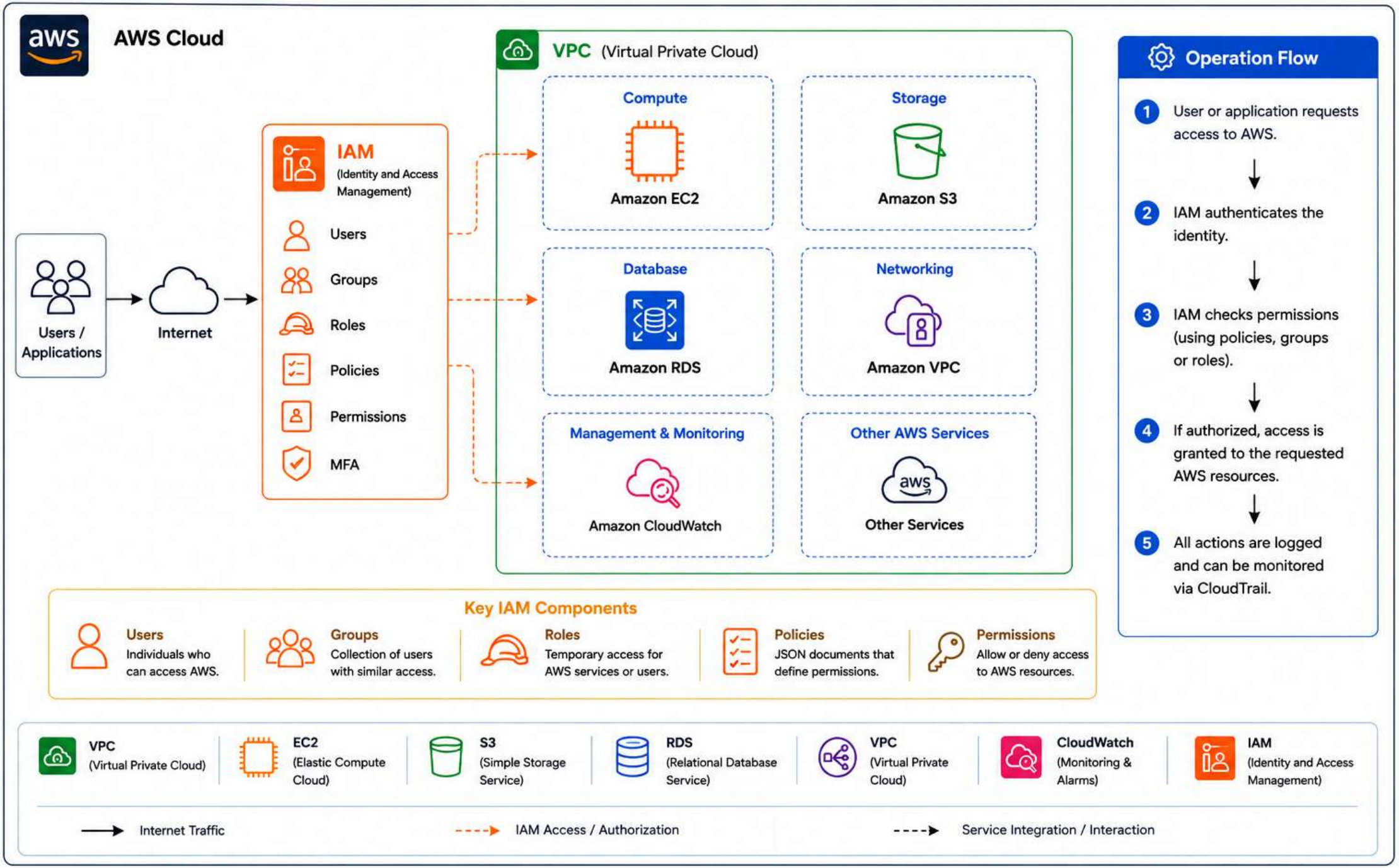
AWS EC2 with Auto Scaling Group



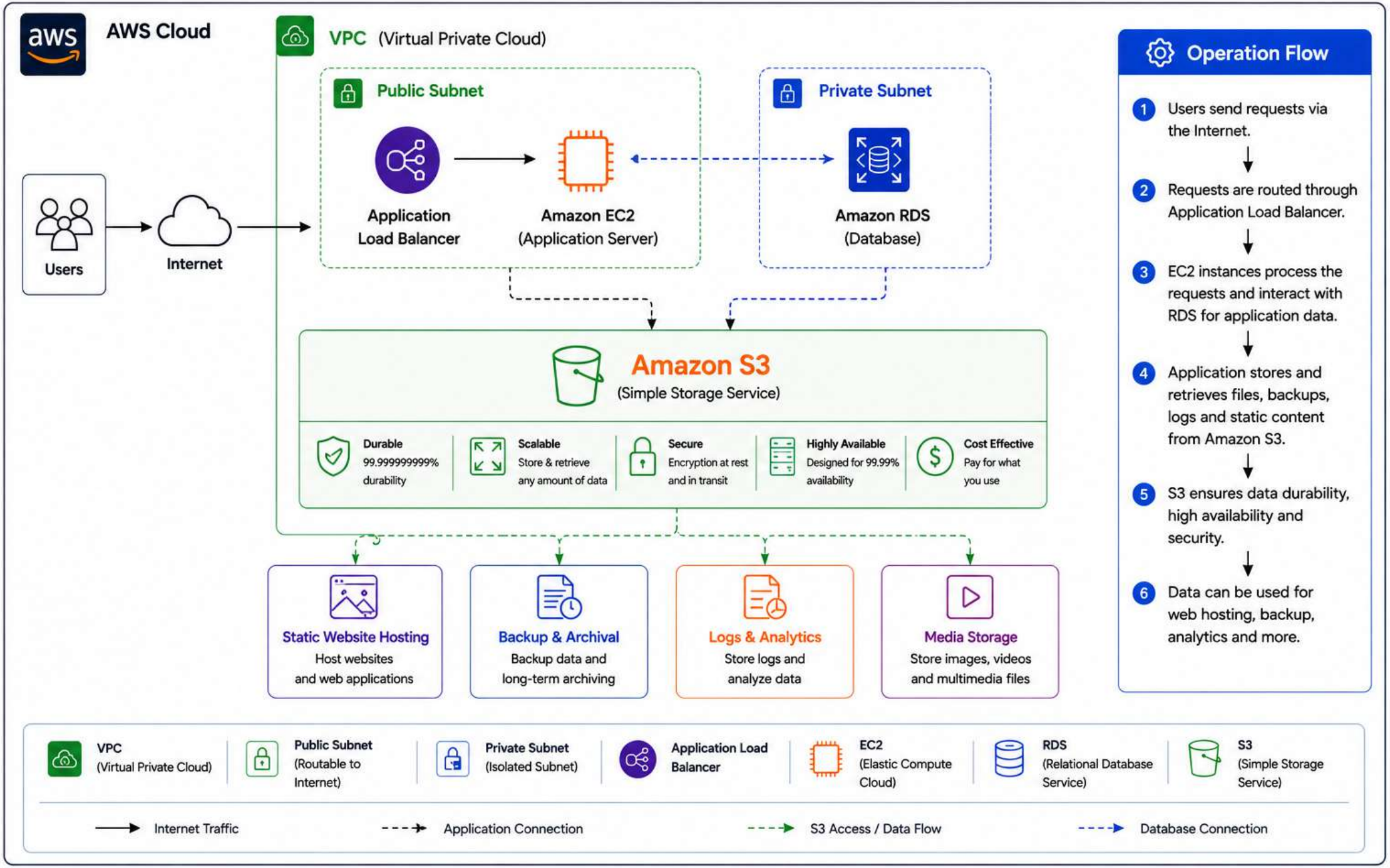
AWS with RDS


AWS Cloud
VPC (Virtual Private Cloud)
Availability Zone A
Availability Zone B
Public Subnet
Public Subnet
EC2 Instance
(Application Server)
EC2 Instance
(Application Server)
Users
Internet
Elastic Load Balancer
(Application Load Balancer)
Private Subnet
Private Subnet
RDS Primary
(Multi-AZ)
Synchronous Replication
RDS Standby
(Multi-AZ)
Amazon S3
(Backups, Logs, Static Assets)
Amazon CloudWatch
(Monitoring & Alerts)
Operation Flow
1 Users send requests via the Internet.
2 Requests go to Elastic Load Balancer.
3 ELB distributes traffic to EC2 instances in multiple Availability Zones.
4 EC2 instances connect to Amazon RDS for database operations.
5 RDS Multi-AZ provides high availability with synchronous replication.
6 Data is backed up to Amazon S3.
7 CloudWatch monitors the resources and sends alerts.
High Availability
RDS Multi-AZ ensures automatic failover and minimal downtime.
Durability
Automated backups and point-in-time recovery.
Scalability
Easily scale compute (EC2) and storage (RDS).
Cost Effective
Pay only for what you use with managed services.
VPC (Virtual Private Cloud)
Public Subnet (Subnet with route to Internet)
Private Subnet (Isolated Subnet)
EC2 Instance (Elastic Compute Cloud)
RDS (Relational Database Service)
Elastic Load Balancer (Application Load Balancer)
S3 (Simple Storage Service)
CloudWatch (Monitoring & Alerts)
Internet Traffic
Application Connection
RDS Replication
Backup / Storage Connection
Monitoring / Logs

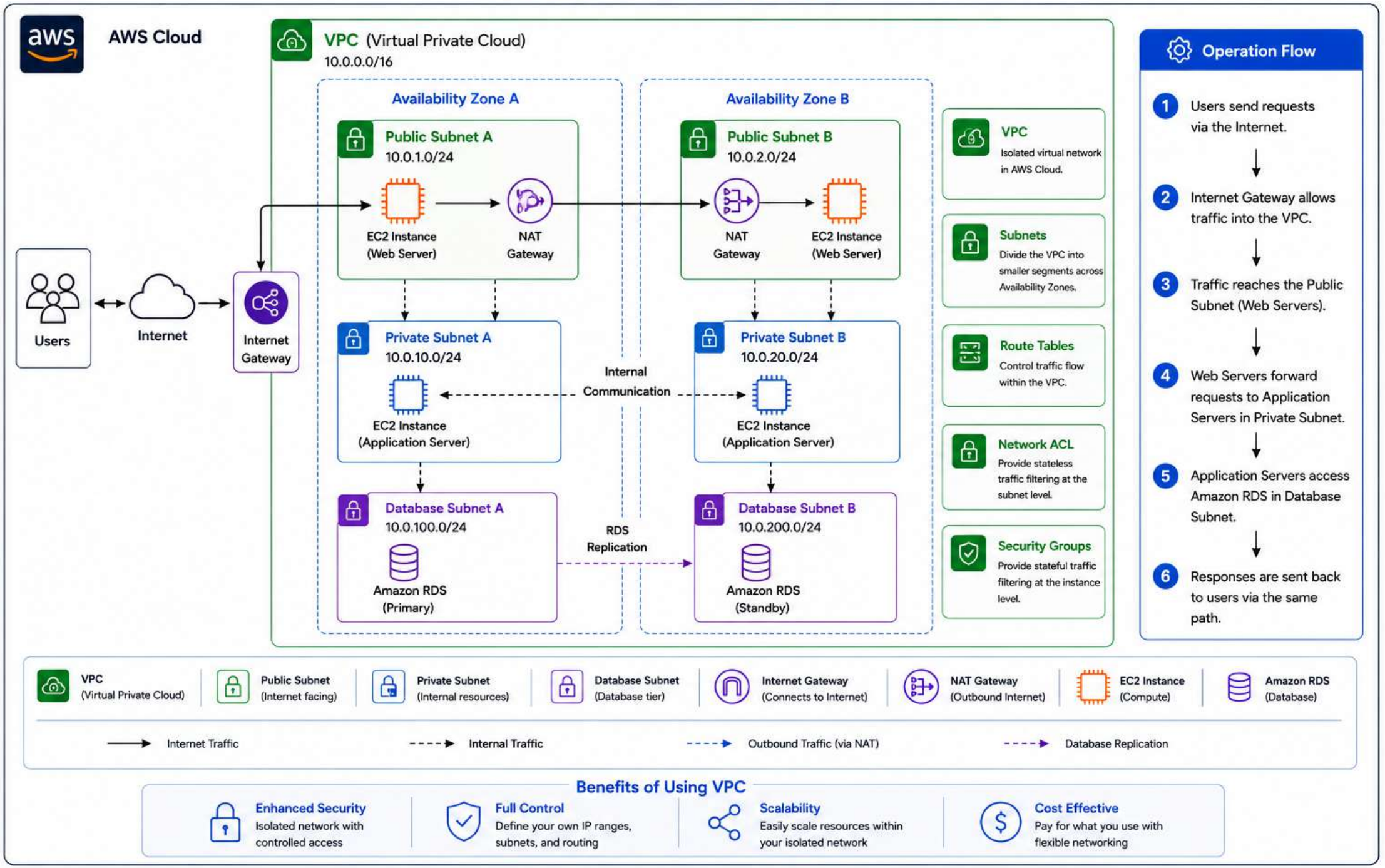
AWS with IAM


AWS Cloud
Users / Applications
Internet
IAM
(Identity and Access Management)
Users
Groups
Roles
Policies
Permissions
MFA
VPC (Virtual Private Cloud)
Compute
Amazon EC2
Storage
Amazon S3
Database
Amazon RDS
Networking
Amazon VPC
Management & Monitoring
Amazon CloudWatch
Other AWS Services
Other Services
Operation Flow
1 User or application requests access to AWS.
2 IAM authenticates the identity.
3 IAM checks permissions (using policies, groups or roles).
4 If authorized, access is granted to the requested AWS resources.
5 All actions are logged and can be monitored via CloudTrail.
Key IAM Components
Users
Individuals who can access AWS.
Groups
Collection of users with similar access.
Roles
Temporary access for AWS services or users.
Policies
JSON documents that define permissions.
Permissions
Allow or deny access to AWS resources.
VPC (Virtual Private Cloud)
EC2 (Elastic Compute Cloud)
S3 (Simple Storage Service)
RDS (Relational Database Service)
VPC (Virtual Private Cloud)
CloudWatch (Monitoring & Alarms)
IAM (Identity and Access Management)
Internet Traffic
IAM Access / Authorization
Service Integration / Interaction

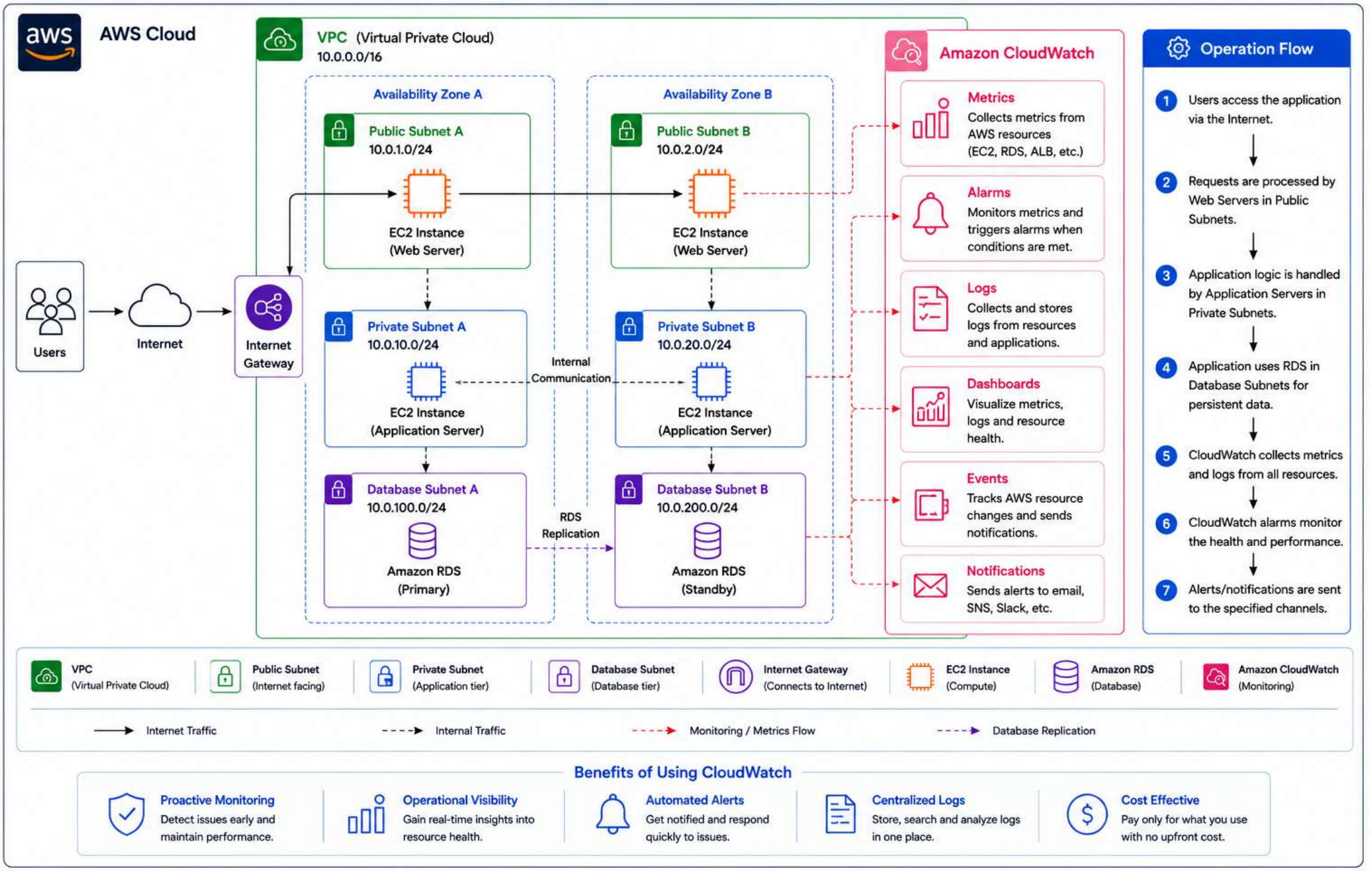
AWS with Amazon S3



AWS with VPC


AWS Cloud
VPC (Virtual Private Cloud)
10.0.0.0/16
Availability Zone A
Availability Zone B
Public Subnet A
10.0.1.0/24
Public Subnet B
10.0.2.0/24
EC2 Instance
(Web Server)
NAT
Gateway
NAT
Gateway
EC2 Instance
(Web Server)
Users
Internet
Internet
Gateway
Private Subnet A
10.0.10.0/24
Private Subnet B
10.0.20.0/24
Internal
Communication
EC2 Instance
(Application Server)
EC2 Instance
(Application Server)
Database Subnet A
10.0.100.0/24
Database Subnet B
10.0.200.0/24
RDS
Replication
Amazon RDS
(Primary)
Amazon RDS
(Standby)
VPC
Isolated virtual network in AWS Cloud.
Subnets
Divide the VPC into smaller segments across Availability Zones.
Route Tables
Control traffic flow within the VPC.
Network ACL
Provide stateless traffic filtering at the subnet level.
Security Groups
Provide stateful traffic filtering at the instance level.
Operation Flow
1 Users send requests via the Internet.
2 Internet Gateway allows traffic into the VPC.
3 Traffic reaches the Public Subnet (Web Servers).
4 Web Servers forward requests to Application Servers in Private Subnet.
5 Application Servers access Amazon RDS in Database Subnet.
6 Responses are sent back to users via the same path.
VPC (Virtual Private Cloud)
Public Subnet (Internet facing)
Private Subnet (Internal resources)
Database Subnet (Database tier)
Internet Gateway (Connects to Internet)
NAT Gateway (Outbound Internet)
EC2 Instance (Compute)
Amazon RDS (Database)
Internet Traffic
Internal Traffic
Outbound Traffic (via NAT)
Database Replication
Benefits of Using VPC
Enhanced Security
Isolated network with controlled access
Full Control
Define your own IP ranges, subnets, and routing
Scalability
Easily scale resources within your isolated network
Cost Effective
Pay for what you use with flexible networking

AWS with Cloud Watch



Benefits of Using CloudWatch

Proactive Monitoring
Detect issues early and maintain performance.

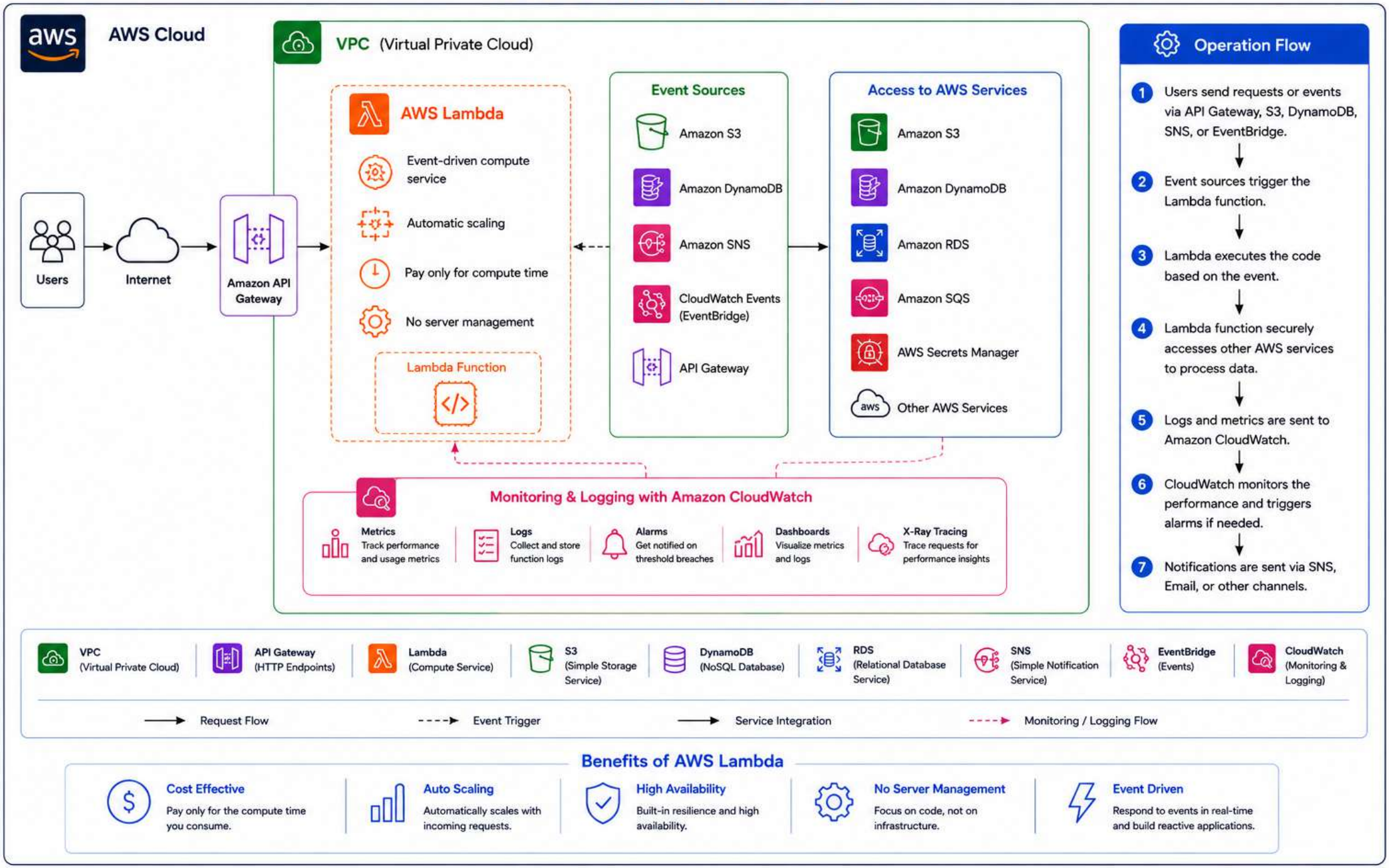
Operational Visibility
Gain real-time insights into resource health.

Automated Alerts
Get notified and respond quickly to issues.

Centralized Logs
Store, search and analyze logs in one place.

Cost Effective
Pay only for what you use with no upfront cost.

AWS LAMBDA



Benefits of AWS Lambda

Cost Effective
Pay only for the compute time you consume.

Auto Scaling
Automatically scales with incoming requests.

High Availability
Built-in resilience and high availability.

No Server Management
Focus on code, not on infrastructure.

Event Driven
Respond to events in real-time and build reactive applications.